Reason Lab README

"Better Learning Starts With You."

Ayala Wang | Anthony Wong | Junjie Liu

Target Audience

College and high school students who are dealing with complex coursework and may be overly reliant on AI shortcuts

Problem

With AI readily available, many students use it to bypass deep reasoning. This shortcut use reduces long-term understanding and reflective learning

Value Proposition

Reason Lab enforces active learning by requiring students to attempt reasoning before receiving assistance. Instead of giving instant answers, the system provides adaptive hints and structured feedback to build durable conceptual understanding

Access & Setup

The medium-fi prototype was built in Figma (link)

1. Open the provided Figma link
2. Click “Present” (top-right corner).
3. Ensure that the device frame is visible.
4. Only blue-highlighted elements are interactive.
5. All interactions are click-based (no real backend).

Intended Device

Reason Lab is designed to run on desktop or laptop computers (Mac, Windows, Linux) using a web browser. Although the interface visually resembles a structured learning dashboard, the intended interaction environment is:

- Large screen workspace
- Keyboard-based typing for explanations
- Mouse/trackpad navigation

  Side menu + extended panels visible comfortably on desktop

Because the system requires users to:

- Type structured explanations,
- Review longer hint responses,
- Import files
- Adjust difficulty sliders,
- Monitor progress bars,

a desktop environment better supports sustained academic work and reflective reasoning compared to mobile devices.

Context of Use

Reason Lab is used when a student:

- Is stuck on a homework problem
- Wants a hint but does not want a full solution
- Wants to reinforce understanding after attempting a problem
- Wants to import material and generate a structured study guide

What Users Should Be Able to Accomplish

- Attempt a problem before receiving assistance
- Receive incremental hints
- Adjust difficulty via slider
- Import files for guided assistance
- Generate a structured study guide
- See visual progress toward receiving a hint

How to Operate the Prototype

Only highlighted elements are clickable.

Task 1 – Simple Flow

Receive an Explanation / Get a Hint

Goal: Demonstrate enforced reasoning before hint release.

Steps:

1. Start at the main problem screen.
2. Click on “Question:” input field to mock an input.
3. In the easiest mode, the system will give hints directly. In harder modes, the system prompts the user to add explanation & thoughts.
4. Enter a reasoning attempt by clicking on “Enter”
5. Press “Submit.”
6. Get hints and concept-level guidance.

Task 2 – Moderate Flow

Import a File

Goal: Demonstrate extended functionality beyond direct problem entry.

Steps:

1. Click on “Import”
2. A file will be added into the input field.
3. The system displays imported content.
4. Press “Submit.”
5. The system allows hint generation from imported files.

Task 3 – Complex Flow

Reach Maximum Hints / Study Guide Generation

Goal: Demonstrate adaptive limit enforcement and reflection.

Steps:

1. Enter a question.
2. Attempt reasoning multiple times.
3. Submit responses.
4. After reaching the hint threshold, the system no longer offers hints.
5. The system restricts additional hints unless the user changes accessibility settings (not implemented in the prototype) OR offers an acceptable explanation of

Hard-Coded Items

Due to medium-fi limitations (Figma), the following are hard-coded:

- Hint text is pre-written and static.
- Maximum hint limit is manually set.
- Slider does not dynamically alter backend reasoning.
- No actual AI model is connected.
- Navigation state resets between sessions.
- No real saving functionality.
- Submit button triggers pre-linked screens only.

Wizard-of-Oz Techniques

- The hint progression is simulated through linked frames.
- The progress visually changes but does not compute reasoning quality.
- Hint limits are manually represented rather than algorithmically enforced.

Limitations

Because this is a medium-fi prototype built in Figma:

1. No real AI processing.
2. No actual reasoning analysis.
3. No dynamic text input and parsing.
4. No real-time evaluation of user explanations.
5. No persistent data storage.
6. Slider is symbolic, not functional.
7. Accessibility options do not compute backend changes.
8. Hint thresholds are simulated.
9. No force hard limits.
10. No actual file imports.
11. Timing and performance metrics cannot be tested.

These limitations were necessary due to:

- Time constraints.
  Focus on interaction design over backend engineering.
- Testing usability of reasoning flow rather than algorithm performance.

Usability Goals

1. Intuitive Navigation

Measured by:

- Navigation errors on repeated tasks.
- Improvement in second attempts.

2. Low Mental Effort (for navigation only)

Measured by:

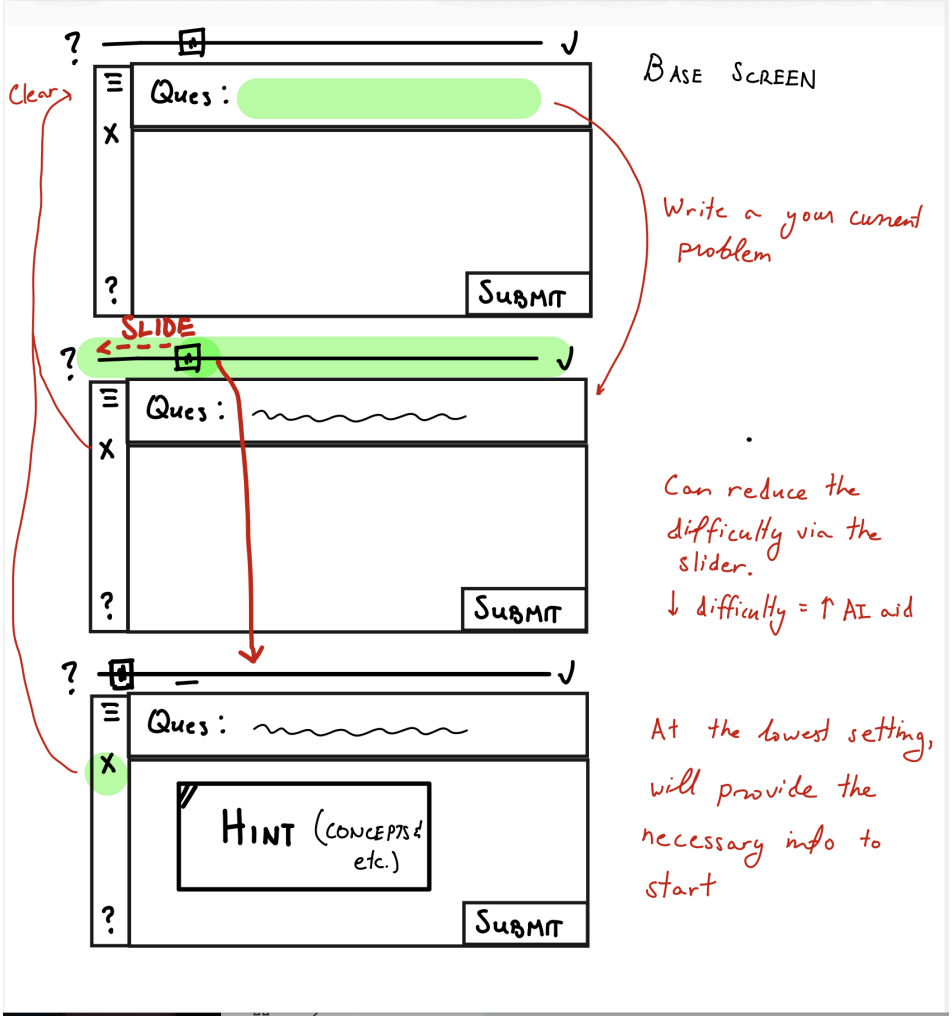
- Self-reported mental effort (1–5 scale).
- Observed hesitation during interface use.

Reason Lab aims to increase cognitive effort in reasoning, not in navigation.

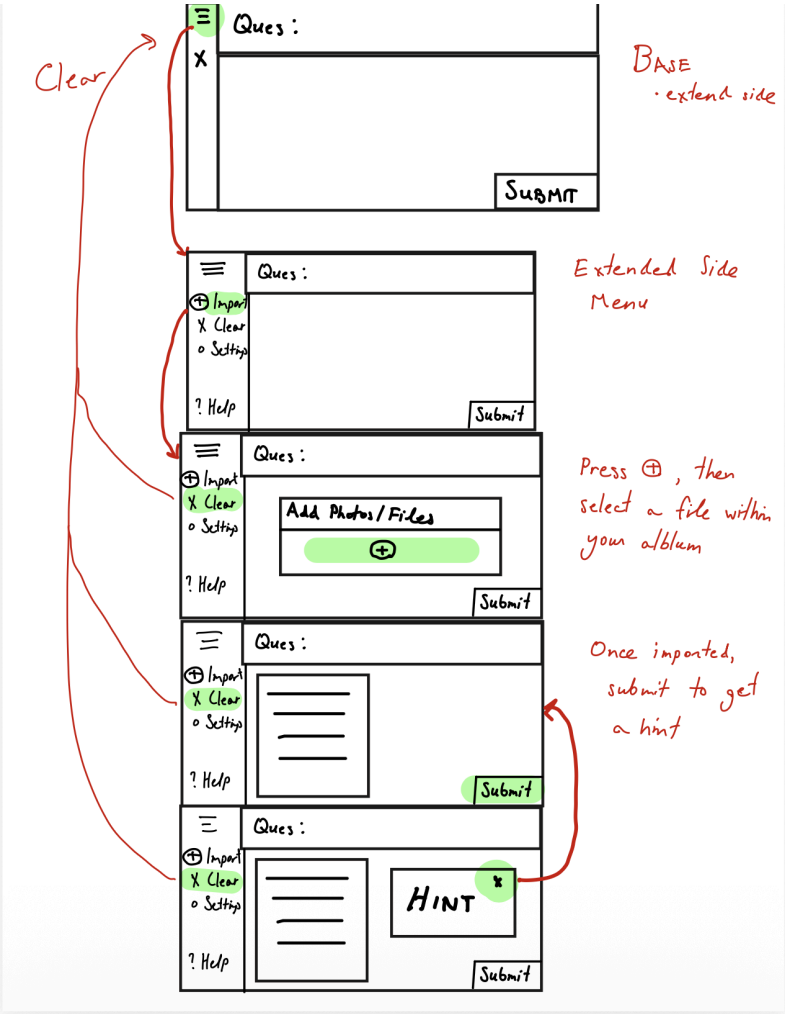
Images

Below are references to key interface visuals from slides

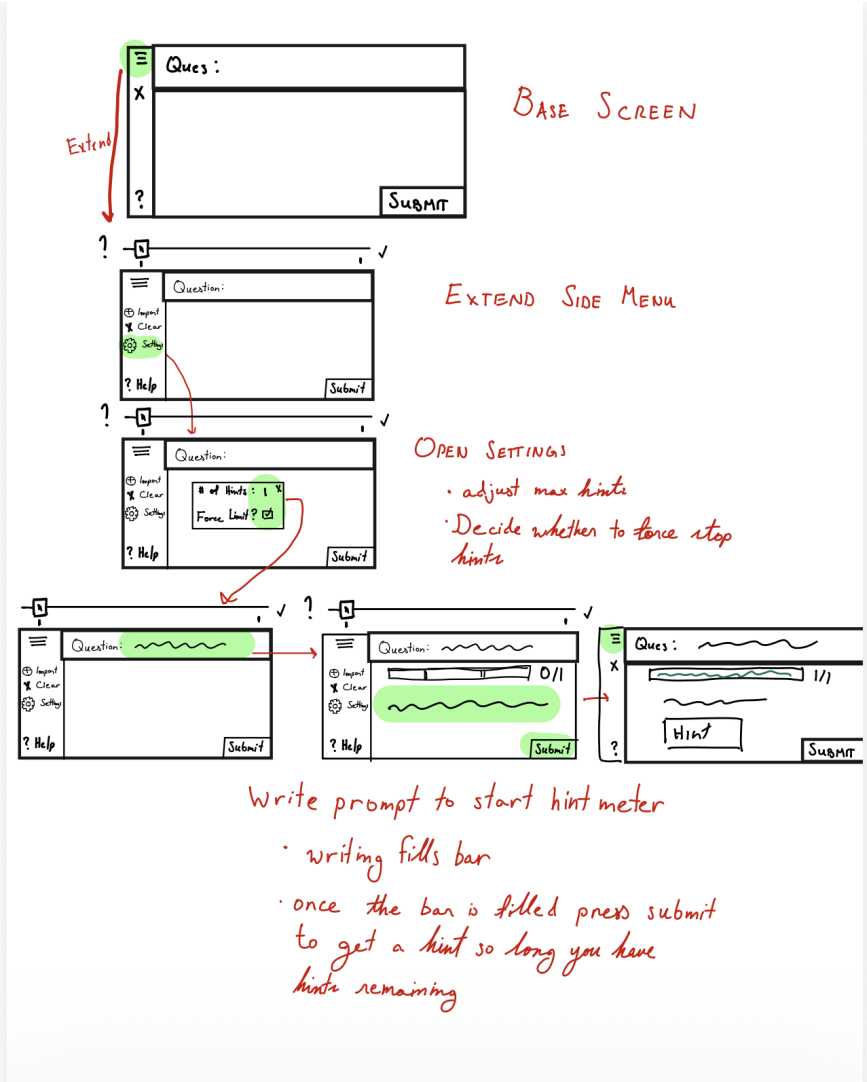
- Task 1 Sketch – page 12



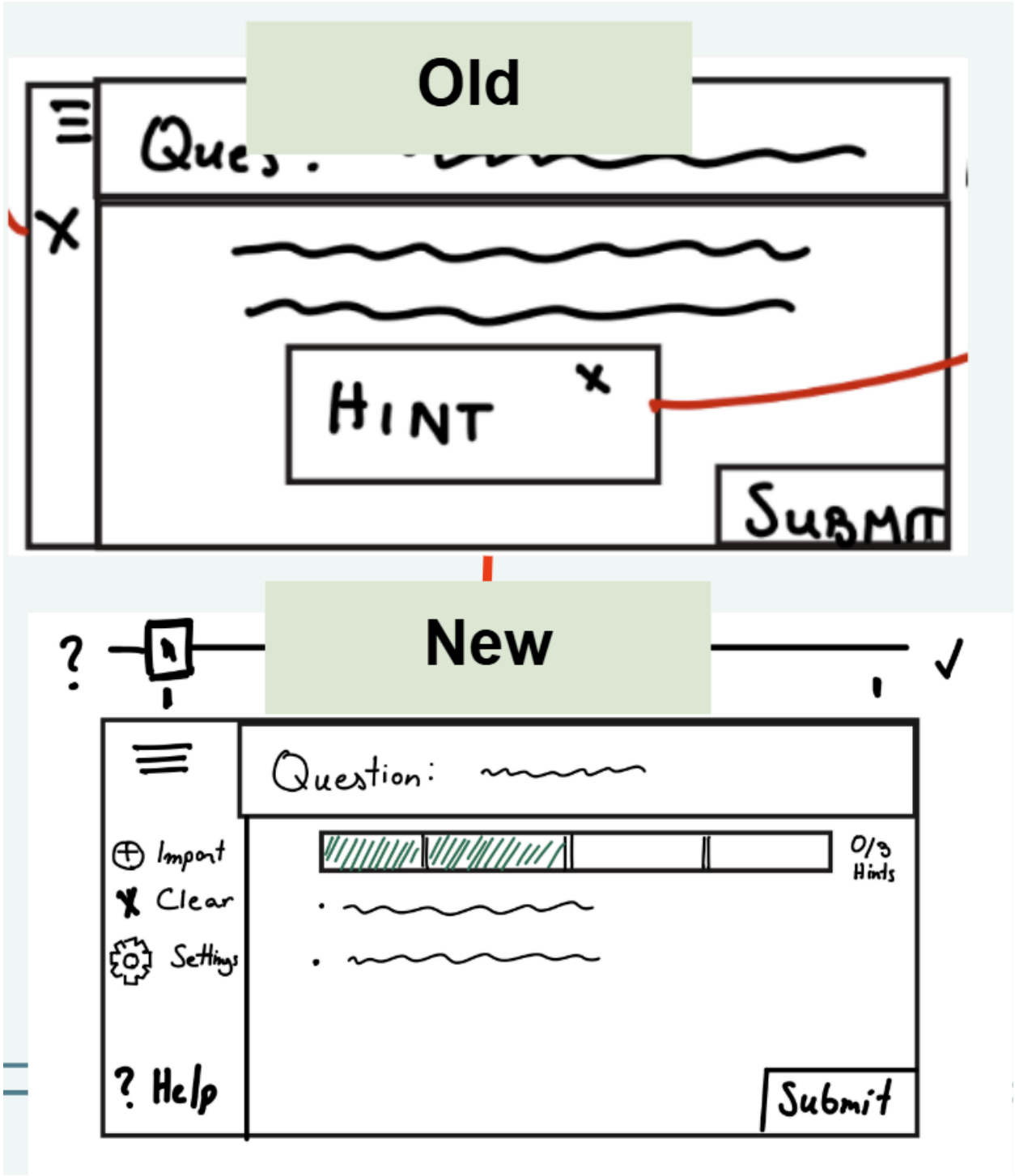
- Task 2 Sketch – page 13



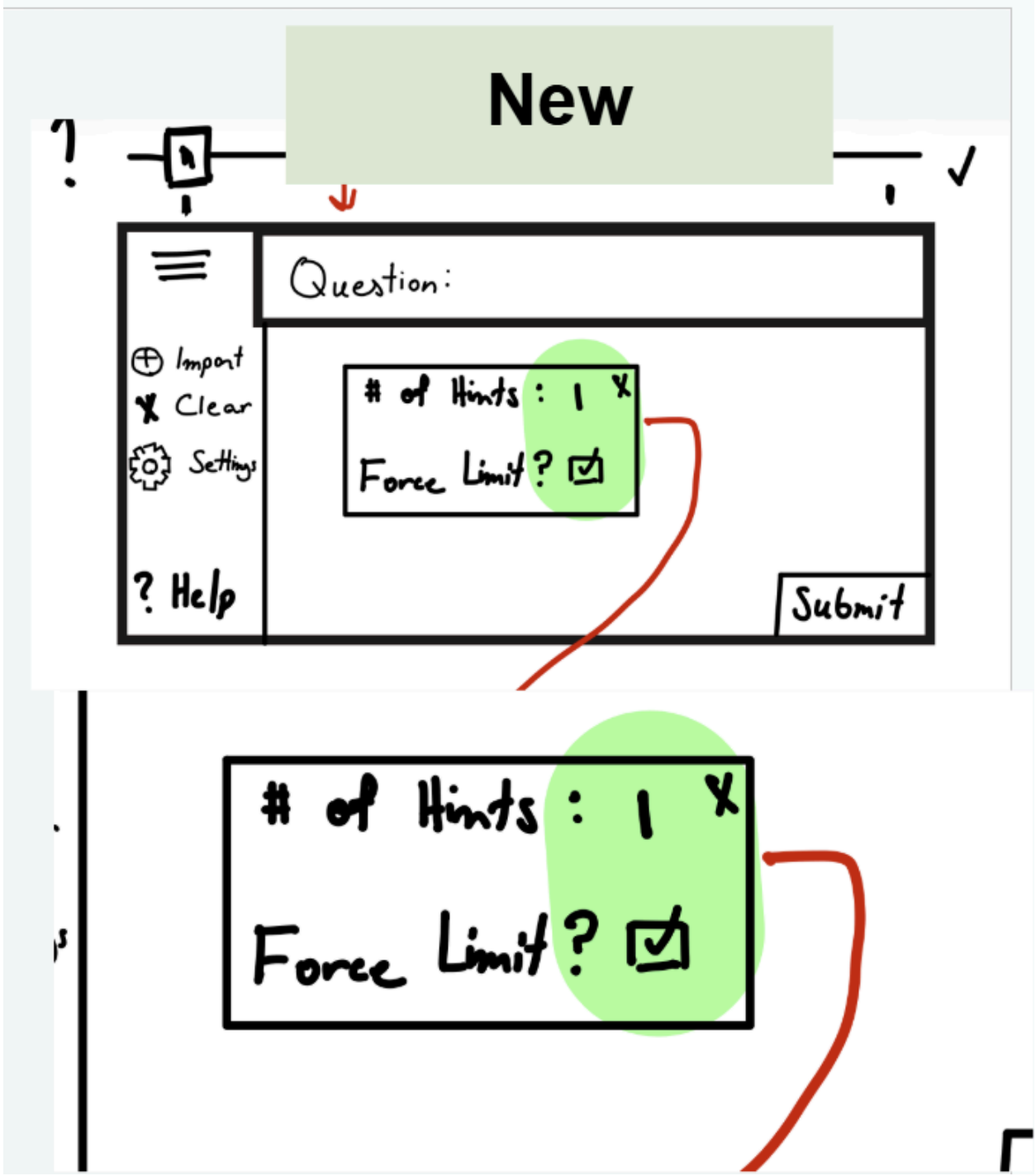
- Task 3 Sketch – page 14



- Progress Bar Change – page 20

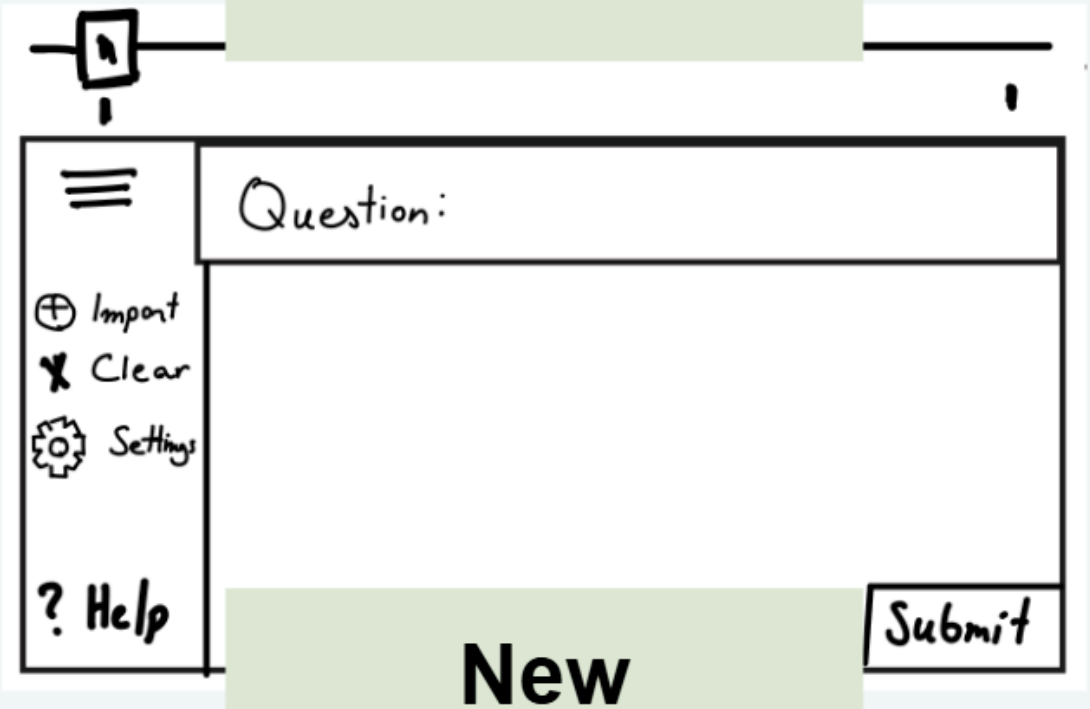


- Accessibility Change – page 22

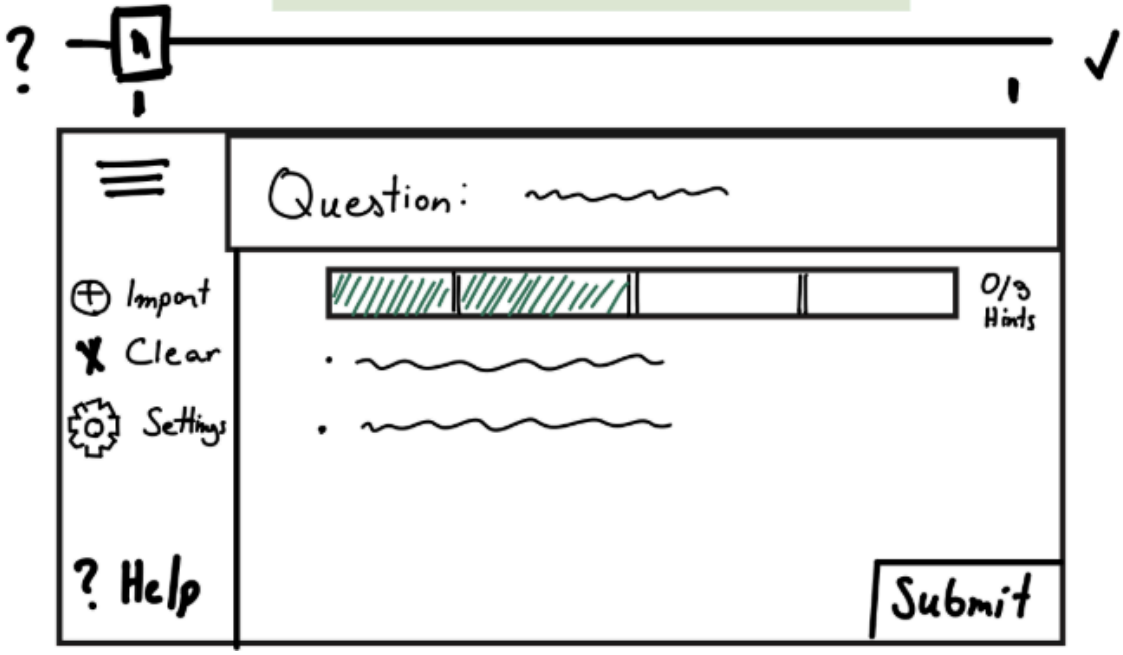


- Slider Redesign – page 24

Old


Question:
Import
Clear
Settings
? Help
Submit

New


?
Question:
Import
Clear
Settings
0/3
Hints
? Help
Submit

Summary

Reason Lab is a learning companion designed to:

- Enforce explanation before assistance
- Limit passive AI dependency
- Provide incremental guidance
- Increase reflective learning
- Maintain low navigation complexity